

MKRT-ELITE CAD FABRICATION GUIDE

V1.3 - UNIVERSAL SOLID-STATE SPEC

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1. Lattice Geometry: Fractal Gyroid

The core uses a nested fractal Schoen Gyroid to handle multiscale kinetic energy (high-frequency buzz to low-frequency impact).

$$\sin(x)\cos(y) + \sin(y)\cos(z) + \sin(z)\cos(x) = 0$$

- **Toolpathing:** **Non-Planar 3D Printing** is mandatory for continuous fiber strength.
- **Lattice Density:** Functional gradient from 10% (Center) to 55% (Shell).

2. Solid-State Reflex Stack

Structural Core:

Conductive CNT-doped PVDF. The lattice acts as the primary circuit for haptic-reflex sensing and self-annealing.

Storage Layer:

Integrated solid-state supercapacitor "Sub-floor" at the baseplate interface. 3mm Z-height.

3. Surface Mineralization & Protection

- **Material:** TPU-h-BN (Hexagonal Boron Nitride) composite skin.
- **Additives:** TiO₂ and ZnO nanoparticles encapsulated for UV protection and thermal reflection.
- **Passive System:** Asymmetric micro-channels for silt trapping and sacrificial mineral scrubbing.

4. Cellular Assembly

- **Shape:** Universal Hexagonal Grid.
- **Interface:** Conductive snap-fit honeycomb grid (Zero-wiring installation).
- **Autonomous Function:** Distributed power sharing across cells via the Hex-Grid Bus.

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